

Question

Write a Java program to demonstrate hierarchical inheritance.

Steps for Execution

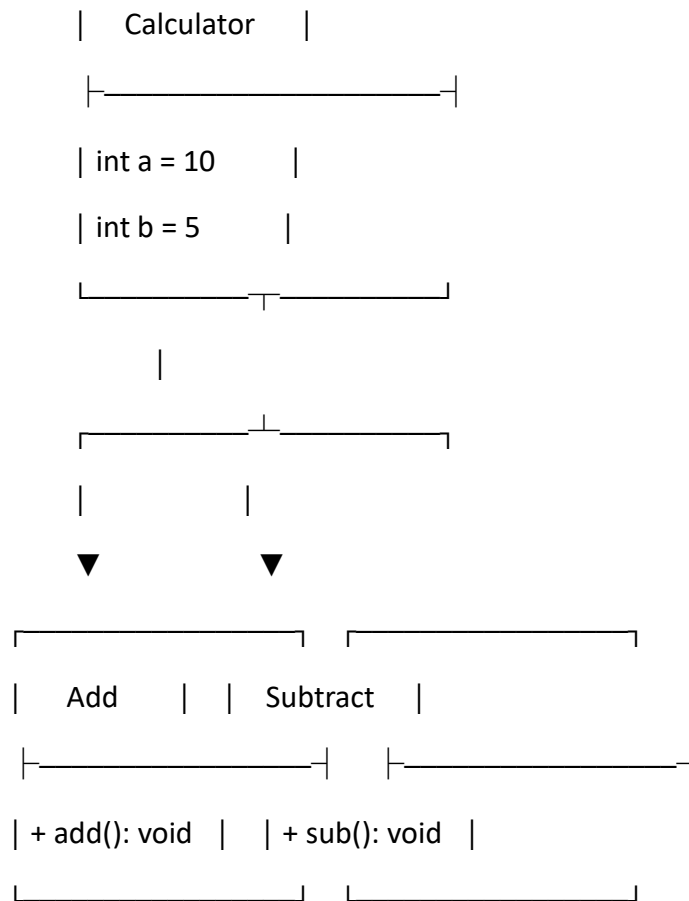
1. Create a package named javaprograms.
2. Create a base class named Calculator.
3. Declare two integer variables $a = 10$ and $b = 5$ in the Calculator class.
4. Create a derived class Add that extends Calculator.
5. Define the add() method in the Add class to perform addition.
6. Create another derived class Subtract that extends Calculator.
7. Define the sub() method in the Subtract class to perform subtraction.
8. Create objects x and y for Add and Subtract.
9. Call x.add() and y.sub().
10. Display the addition and subtraction results.
11. Stop the program.

Algorithm

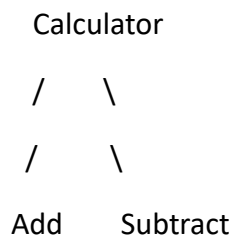
1. **Start**
2. Create a superclass Calculator.
3. Initialize $a = 10$ and $b = 5$.
4. Create subclass Add extending Calculator.
5. Perform $a + b$ in the add() method.
6. Create subclass Subtract extending Calculator.
7. Perform $a - b$ in the sub() method.
8. Create objects of both subclasses.
9. Call the respective methods.
10. Display the results.
11. **Stop**

UML Class Diagram





Inheritance structure:



This is called **hierarchical inheritance** because multiple subclasses (Add and Subtract) inherit from the same superclass (Calculator).

Code

Hierarchical Inheritance Java Program

```
package javaprograms;
```

```
class Calculator {
    int a = 10, b = 5;
}
```

```
class Add extends Calculator {  
    void add() {  
        System.out.println("Addition = " + (a + b));  
    }  
}
```

```
class Subtract extends Calculator {  
    void sub() {  
        System.out.println("Subtraction = " + (a - b));  
    }  
}
```

```
public class Hierarchical {  
    public static void main(String[] args) {  
        Add x = new Add();  
        Subtract y = new Subtract();  
  
        x.add();  
        y.sub();  
    }  
}
```

Output

Addition = 15

Subtraction = 5

Result

The program successfully demonstrates **hierarchical inheritance** in Java, where two subclasses, Add and Subtract, inherit the properties of the common superclass Calculator.

